

General Science Style

DICTIONARIES AND REFERENCE MATERIAL

Editions of style manuals and dictionaries to be used (the order in which they are listed here does not indicate preference):

Webster's Third New International Dictionary

American Chemical Society Style Guide

Chicago Manual of Style

Council of Science Editors Style Manual

ASM Style Manual (American Society for Microbiology, 1991) (This style manual is no longer being published, but existing copies are good sources for biology style information.)

Stedman's Medical Dictionary

The Elements of Style, Strunk and White

For a question about a specific term, an editor or proofreader should first consult the *Science* style manual, then *Webster's*, and then the appropriate reference listed above. Editors should note, however, that *Science* style makes some exceptions in specific cases to following the style of certain dictionaries (Wilms tumor, for example). Glossaries of terms may also be consulted. It should be noted that there may be conflict in the way various authorities treat terms. For example, *Webster's Third* gives "wave form"; the American Institute of Physics and McGraw-Hill give "waveform." We use "waveform" in physical science papers.

GENERAL STYLE CONCERNS

—Avoid overediting. Try to retain the author's language as much as possible, no matter how idiosyncratic, once it is past the pre-edit stage.

—Because *Science* papers report only new developments, claims that studies are being done "for the first time" or that the work is "new," "novel," or "unique" are not necessary and should be removed or recast. In addition, references to work that is to follow or general statements about what is "in the literature" or "published" that lack specific references should also be removed.

—Use the past tense to report experimental results and the present tense to state conclusions. Specific references to the work of earlier investigators should be in the past or past perfect tense.

—Active voice is more appropriate than passive voice in cases in which an author's procedures need to be distinguished from previous work. But editors should make sure not to overedit in changing from passive to active voice. In reporting results, the results should be emphasized, not the authors.

—Avoid laboratory jargon specific to certain disciplines (i.e., "osmicated").

–Do not use italics, boldface, or any special font or typeface for emphasis. The author's words should be sufficient without special emphasis. If the author coins a new term or uses a word in a special sense, it is okay to use quotes around the new word at the first usage; quotes are usually not retained in subsequent usages of the word unless the word is meant in a special sense. Boldface in figure legends should be used for figure panels (**A**, **B**, etc.) or to indicate location (**left**, **right**, **top**, **bottom**, etc.) if the figure has no labeled panels. The first sentence of each figure legend (the figure “title”) should also be in bold. (See the separate section on figure style for details.)

–All figure and table legends should have titles (i.e., bold first sentences) for all *Science* content. Titles should be brief descriptions of the content of the figure or table.

–We use bold and italic for different levels of subheading. We do not allow a subhead (e.g., “Introduction”) between the abstract and the first paragraph. First-level subheads are for major topic changes. When a secondary division is needed, there should be at least two subheads within a division. Reports do not have subheads. For all *Science* content, all subheads should be on separate lines, to be styled as follows:

–Level 1 heads: Sentence case and bold (Example: “**A perspective on genome-based medicine in oncology**”)

–Level 2 heads: Sentence case, bold, and italic (Example: “***Opportunities in genome-based medicine***”). Italic terms (such as gene names) mentioned within these subheads should be roman. Note that in the HTML version of the paper, these heads are title case, grayscale, and not italicized.

–Level 3 heads (rarely used): Sentence case and italic (Example “*Further opportunities*”)

When a second-level head falls directly after a first-level head (stacked headings), both are set flush left.

–Use SI or metric units of measure. Nonmetric units may be used if the original measurement was made in nonmetric units. If appropriate, a conversion into metric units may be inserted for clarity.

Ex: a 1-inch pipe

The diameter was 1 inch (2.5 cm).

–Spell out all acronyms and abbreviations the first time they are mentioned in the text. Consult the *Science* Acronyms list to see which ones are spelled out per *Science* style.

– Use U.S. English spellings only, except in author affiliations, organizations, journal and book titles, and place names that do not follow U.S. spellings.

–Write out genus and species names at first use and abbreviate the genus thereafter. (“*Escherichia coli* were found to grow well under these conditions. In later experiments with *E. coli*, . . .”). It is OK to begin a sentence with the abbreviated genus name.

–Try to avoid starting a sentence with an acronym, especially an acronym that begins with a lowercase letter (cDNA). Editors should recast the sentence, if possible.

–Information in the abstract should not be directly repeated in the text.

–If "data not shown" appears in the text, it should (i) be deleted, and the statement accompanying it should be left as an assertion if it is a control or variation on an experiment already described in the paper, or (ii) be replaced with a reference to the supplementary materials and the data inserted into the supplementary materials. In figures, the phrase "not shown" may be used if it means "not shown in the figure."

–Use a solidus (/) only for ratios, division, compound units, particular designations in genetics nomenclature, solvent designations (e.g., methanol/1-propanol), layered materials, and "and/or."

–Use nonsexist language (an editor or proofreader should watch his or her pronouns, not just his).

–Spell out US state names in running text, both when standing alone and when following the name of a city. However, the abbreviation "DC" should always be used (as in "Washington, DC"). Also, states may be abbreviated when used parenthetically to denote a company location; e.g., "(catalog no. 156847, Genentech, San Francisco, CA)".

LANGUAGE USE

I. Wordiness and redundancy

–Editors should watch for common mistakes of wordiness.

Change: The natural oil seepages . . . were the first signs that pointed to . . .
To: The natural oil seepages . . . were the first signs of . . .

Change: To fully appreciate
To: To appreciate

–Delete unnecessary adjectives and adverbs, such as very precise, quite, rather, interesting, remarkable, simply, and clearly. Be careful not to delete adjectives or adverbs that help convey shades of meaning and comparison. For example, do not delete "very" in a construction such as "The results of Jones are very precise in relation to those of Smith."

–Avoid overuse of "there are," and "it is." Almost always delete "note that" and other such instructions to the reader.

Change: There are many studies to substantiate this view.
To: Many studies substantiate this view.

II. Clarity

–Avoid "freight-train" modifiers. Rearrange the sentence, inserting prepositions to clarify what modifies what.

Change: A well-stocked, 40-m-deep freshwater California lake
To: A freshwater lake in California that is 40 m deep and well stocked . . .

–Provide a noun after "this" when the antecedent is unclear.

Change: This shows . . .
To: This technique shows . . .
 result
 model
 approach

–Editors should be mindful of metaphors and figures of speech.

"Shelving legislation" is OK, does not arouse inaccurate image
"Shelving plans" is reasonable
"Shelving construction" is absurd!
"Psychotherapy research is being pushed out of the nest . . ."
"The subpoena catapulted the case from routine to . . ."

–Editors should be careful to compare like with like.

Incorrect: Jones compared the profiles for his group with age-matched controls.

Correct: Jones compared the profiles of his group with those of age-matched controls.

–Reword dangling modifiers, such as participles and infinitives that do not have a logical or clear reference in the sentence. Editors and proofreaders should be aware that this is a particular problem with the passive voice.

Incorrect: To verify the accuracy of the results, the experiment was repeated.
[who was doing that?]

Correct: To verify the accuracy of the results, we repeated the experiment.

Incorrect: Using the same procedure as before, the report concluded . . . [who was doing that?]

Correct: Using the same procedure as before, the researchers concluded that . . .

PARTS OF A PAPER

I. Affiliations and Titles

–Affiliations should give a complete address for each author. Editors should make sure to indicate which author is in which department if several departments are given. The address given should correspond to the place or places where the work was done. If an author has moved since completing the work, the affiliations should give additionally his or her present address.

–See the Author Affiliations section of the *Science* Style guide for detailed information on affiliation style.

–Foreign language affiliations need not be translated to English.

–Organizations that follow British spelling conventions should retain their British spelling.

Ex: Global Programme on AIDS

–Information on the author who will receive correspondence (if not by email), authors on sabbatical, and present address information should be given in the form of footnotes. The corresponding author's address should be precise enough that mail can be delivered.

–Affiliations of authors should be listed in the order of authorship.

–Affiliations in Reports are in the form of addresses. Affiliations in Perspectives and Policy Forms are in complete sentences [not always; for multiple authors, we have been using superscript numbers and footnote-style addresses].

- Affiliations should include ZIP Code (if provided) and USA after all of the U.S. affiliations in the following format: Stanford, CA 94305, USA. If the author does not provide a ZIP code and the address is otherwise clear, the ZIP code can be omitted (but preferable to be internally consistent with omission versus inclusion of ZIP codes within a given paper). Postal codes are not required for non-U.S. addresses.
- Titles can be either declarative sentences or descriptive phrases. Titles cannot exceed 96 characters and spaces total; four-line titles are not allowed.

II. Text

- Paragraphs in the text should contain at least two sentences and be no longer than a typed page.
- Avoid a summary or restatement of the preceding information in the last paragraph of the paper. This paragraph may contain concluding remarks only. Avoid speculative language.
- Symbols as verbs in a sentence in the text should be changed to words (do not use = in place of "is" or "equals" when that is the verb of the sentence).
- To name authors in the text, it is common to use only the name of the first researcher and *et al.* or co-workers, colleagues, etc., if there is more than one author (i.e., “Smith and co-workers discovered that...” or “Smith *et al.* discovered that...”) Make sure that there is a reference number fairly close so that it is clear where the reader can find Smith *et al.*’s work. It is up to the author what particular word (“co-worker,” “colleague,” etc.) is used; query the author if a usage seems unclear. Alternatively, the author may name a researcher in the text whose name is not the first in the reference (i.e., Smith is the fourth author) but at whose laboratory the work was done. In these cases, both “co-workers” and “colleagues” are acceptable terms, but do not say “Smith *et al.*...” or “Smith and others...”

–How to do block quotes

For quoted material long enough to warrant that the quotation be set off, use the following format:

- 1) no quotation marks around the material
- 2) mark quote to be set in 8.5-pt type
- 3) mark 8-pt space above the quote and 6-pt space below the quote
- 4) a page number should appear after the reference number

It is not necessary to verify a quote with its original source, except for ones taken from *Science*. If a quoted item does not seem accurate, query the author. For punctuation after a nonblock quote, see the punctuation section. For block quotes in a note, do all of the above but do not set the type smaller.

–How to do series

All parts in a series, whether itemized or not, should be parallel in construction (all phrases or all sentences, not a mixture).

A. Non-itemized Series

Usually, series that are not itemized do not take a colon, particularly if the series can be introduced by an expression such as "for instance," "namely," or "that is."

Incorrect: The metals excluded were: molybdenum, mercury, and magnesium.

Correct: The metals excluded were molybdenum, mercury, and magnesium.
(from *Chicago*)

Incorrect: The study did not account for many metals: molybdenum, mercury, and magnesium.

Correct: The study did not account for many metals—for example, molybdenum, mercury, and magnesium.

However, a list when set off can be introduced by a colon, particularly if it is a grammatically complete clause.

Ex: This study included three areas: McBurney Point, Rockland, and Effingham.

For example: Morton had praised poodles for many years; Gilbert disliked poodles intensely; Gilbert and Morton seldom looked each other in the eye. (from *Chicago*)

For more information, see *Chicago*.

B. Itemized series

1. Series within a paragraph. Use lowercase i, ii, iii, iv, . . . enclosed in parentheses.

Ex: Possible disturbances include (i) silicate contamination and (ii) diffusion loss. There are several ways to use itemized series within a paragraph.

Complete sentence: (i) fragment, (ii) frag., and (iii) frag.

Complete sentence. [or :] (i) Sentence(s). (ii) Sentence(s). (iii) Sentence(s). [no "and"; a period or colon may be used after the initial complete sentence, depending on how the sentence that follows is meant].

Complete sentence. (i) Statement. Sentence(s). (ii) Statement. Sentence(s). (iii) Statement. Sentence(s). [no "and"; this form works best when the statements are very short, perhaps one or two words only]

2. Series in different paragraphs.

Example:

. . . The most important features of the expenditure trends should be considered.
1) Spending is now highly concentrated in a few programs. . . [continues with one paragraph worth of information].
2) Mandatory programs have grown much faster than discretionary programs. . . [one paragraph].
3) Benefits payments in these mandatory programs in the aggregate go predominantly to high- and middle-income beneficiaries. . . [one paragraph].

Each item numbered in such a series is indented as a paragraph and has enough information to make a full paragraph by itself.

III. References, Notes, and Acknowledgments

–Individual notes should not be longer than a typed page.

–If there are any acknowledgments, they should be the last (unnumbered) note and should not be cited anywhere in the paper.

–Do not use *op. cit.* or *ibid.* in references.

–All references should be cited in numerical order. Renumber and rearrange as necessary. Consecutive reference numbering should go in this order: text, text boxes, figure and table legends, reference notes and acknowledgments, and SM. The numbering can run through either the figure or table legends first, but it should be consistent within a paper. Every reference cited separately should have its own reference number. If a particular citation in a group of references is cited separately, it will be necessary to split these references and renumber. Each reference should appear only once in the reference set.

–References that appear only in the supplementary materials should be included in the main reference list and cited in numerical order, starting directly after the last main-text reference entry.

–Notes added in proof, usually noting something that the authors have just been made aware of, should be handled carefully. Copyeditors should make the editor in charge of the paper aware of a note added in proof, no matter when it is added by the author. In general, if such a note is added before the paper is accepted, it should be incorporated into the text. If it is added after the paper has been accepted but before it has gone to the printer, it is up to the editor in charge of the paper to decide whether the note added in proof is made part of the text or is retained as a note added in proof. Any note added in

proof put in after the paper has gone to the printer should be made a note added in proof on the galleys.

–Style for note added in proof: It should be set as the last paragraph of text and indented with the words "*Note added in proof:*" in italics, followed by the text of the note added in proof.

Ex.: *Note added in proof:* Text of note follows. Text of note follows. Text of note follows. Text of note follows. Text of note follows. Text of note follows. Text of note follows (XX).

Any new reference or references that the note added in proof refers to are placed in the reference list in consecutive order after the last main text reference; all subsequent references and callouts will need to be renumbered accordingly, including SM-only references.

– Avoid paragraph breaks in individual notes.

–As of 4/8/2016, we will allow dedications, but only when the subject of the dedication is deceased. So, for instance, “This work is dedicated in memory of [name here]” is acceptable, whereas “This work is dedicated to [name here], on the event of his/her 50th birthday” is still not allowed. (Of course, if the dedication is being made in honor of a deceased person’s birthday, that would be okay.) If the nature of a dedication is unclear, query the authors and/or editor. Dedication statements are permitted only in the Acknowledgments (and not as footnotes).